Introduction to GLMs

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1 Introduction

Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
        # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

2 Welcome

Welcome to Epidemiology 204: Quantitative Epidemiology III (Statistical Models).

Epi 204 is a course on **regression modeling**.

3 What you should already know

> ⚠ Warning
>
> Epi 202, Epi 203, and Sta 108 are prerequisites for this course. If you haven't passed one of these courses, talk to me ASAP.

3.0.1 Epi 202: probability models

- Probability distributions
  - binomial
  - Poisson
  - Gaussian
  - exponential

- Characteristics of probability distributions
  - Mean, median, mode, quantiles
  - Variance, standard deviation, overdispersion

- Characteristics of samples
  - independence, dependence, covariance, correlation
  - ranks, order statistics
  - identical vs nonidentical distribution (homogeneity vs heterogeneity)
  - Laws of Large Numbers
  - Central Limit Theorem for the mean of an iid sample

3.0.2 Epi 203: inference for one or several homogenous populations

- the maximum likelihood inference framework:
  - likelihood functions
  - log-likelihood functions
  - score functions
  - estimating equations
  - information matrices
  - point estimates
  - standard errors
  - confidence intervals
  - hypothesis tests
  - p-values

- Hypothesis tests for one, two, and >2 groups:
  - t-tests/ANOVA for Gaussian models
  - chi-square tests for binomial and Poisson models
  - nonparametric tests:
    - Wilcoxon signed-rank test for matched pairs
    - Mann–Whitney/Kruskal-Wallis rank sum test for ≥ 2 independent samples
    - Fisher's exact test for contingency tables
    - Cochran–Mantel–Haenszel-Cox log-rank test

For all of these prerequisite concepts and quantities, and especially for confidence intervals and p-values, you should know how **both**:

- how to compute them
- how to interpret them

3.0.3 Stat 108: linear regression models

- building models for Gaussian outcomes
  - multiple predictors
  - interactions
- regression diagnostics
- fundamentals of R programming; e.g.:
  - Wickham et al. (2023)
  - Dalgaard (2008)
- RMarkdown or Quarto for formatting homework[1]
  - LaTeX for writing math in RMarkdown/Quarto

4 What we will cover in this course

- Linear (Gaussian) regression models (review and more details)
- Regression models for non-Gaussian outcomes
  - binary
  - count
  - time to event
- Statistical analysis using R

We will start where Epi 203 left off: with linear regression models.

The type of the outcome variable Y largely determines which regression model we use (Table 1).

Table 1: Outcome types, examples, and the regression analyses that match them

Outcome	Examples	Analysis
binary	diseased/not; died/survived	logistic regression
count	number of infections per month; number of follow-up visits per patient	Poisson regression
time to event	time from exposure until disease onset; time from onset until recurrence; time from disease onset until disease progression; time from onset until death	survival analysis
ordered levels[2]	pain level (0 to 10 scale); approval rating (0 to 5 stars)	ordinal regression

5 The scientific method and models

As scientists, we attempt to answer factual questions about the world and universe in the past, present, and/or future. In this course, we follow the steps of the scientific method[3], with a statistical model standing in for the hypothesis:

- conceive a question
- hypothesize a specific version of that model
- propose a study design that would provide data relevant to the model

[1] https://r4ds.hadley.nz/quarto
[2] Time permitting.
[3] https://en.wikipedia.org/wiki/Scientific_method

- analyze the data
- estimate model parameters
- "inference":
  - quantify uncertainty
  - draw conclusions

Regression models are the family of models we use for these steps.

6 Motivations for regression models

Exercise 6.1. Why do we need regression models?

Solution

Solution 6.1.

- when there's not enough data to analyze every subgroup of interest individually
- especially when subgroups are defined using continuous predictors

---

6.1 Uses of regression models

Vittinghoff et al. (2012, sec. 1.3) identifies three broad motivations for using multipredictor regression models:

> Multipredictor regression can be a powerful tool for addressing three important practical questions. … [These] include prediction, isolating the effect of a single predictor, and understanding multiple predictors.
>
> 1. **Prediction**: "Multipredictor regression is a powerful and general tool for using multiple measured predictors to make useful predictions for future observations."
> 2. **Isolating the Effect of a Single Predictor**: "In settings where multiple, related predictors contribute to study outcomes, it will be important to consider multiple predictors even when a single predictor is of interest." (e.g., to minimize confounding and support causal interpretation)
> 3. **Understanding Multiple Predictors**: "Multipredictor regression can also be used when our aim is to identify multiple independent predictors of a study outcome — independent in the sense that they appear to have an effect over and above other measured variables." (including mediation and interaction)

Kleinbaum et al. (2014, sec. 4.1) provides a more granular list of eight overlapping applications of regression analysis:

> In practice, a regression analysis is appropriate for several possibly overlapping situations, including the following:
>
> 1. **Characterize the association**: "You want to characterize the relationship between the dependent and independent variables by determining the extent, direction, and strength of the association."
> 2. **Prediction**: "You seek a quantitative formula or equation to describe (e.g., predict) the dependent variable Y as a function of the independent variables $X_1, X_2, \dots, X_k$."
> 3. **Controlled description**: "You want to describe quantitatively or qualitatively the relationship between $X_1, X_2, \dots, X_k$ and Y but control for the effects of still other variables $X_{k+1}, X_{k+2}, \dots, X_{k+p}$, which you believe have an important relationship with the dependent variable."
> 4. **Variable selection**: "You want to determine which of several independent variables are important and which are not for describing or predicting a dependent variable.

You may want to control for other variables. You may also want to rank independent variables in their order of importance."

5. **Model selection**: "You want to determine the best mathematical model for describing the relationship between a dependent variable and one or more independent variables."
6. **Comparing regression relationships**: "You want to compare several derived regression relationships." (e.g., whether a relationship between two variables differs across subgroups)
7. **Interaction**: "You want to assess the interactive effects of two or more independent variables with regard to a dependent variable."
8. **Adjusted coefficient estimation**: "You want to obtain a valid and precise estimate of one or more regression coefficients from a larger set of regression coefficients in a given model." (i.e., estimating the effect of one variable after adjusting for others)

6.1.1 Relating the two lists

The two lists use different levels of granularity to describe the same landscape of regression uses. Vittinghoff et al. (2012) provides three broad categories, while Kleinbaum et al. (2014) identifies eight more specific applications.

Table 2: Comparing the categorizations of regression model uses in Vittinghoff et al. (2012, sec. 1.3) and Kleinbaum et al. (2014, sec. 4.1)

Vittinghoff et al. (2012) category	Kleinbaum et al. (2014) application(s)
Prediction	Application 2 (prediction)
Isolating the Effect of a Single Predictor	Application 8 (adjusted coefficient estimation)
Understanding Multiple Predictors	Application 1 (characterize association), Application 3 (controlled description), Application 4 (variable selection), Application 5 (model selection), Application 6 (comparing regression relationships), Application 7 (interaction)

Applications 6 and 7 are related but distinct: Application 6 asks whether a derived regression relationship (e.g., a coefficient or the overall model) differs across pre-defined groups, typically by comparing models fit separately for each group. Application 7 asks whether two predictors interact within a single model — that is, whether the effect of one predictor on the outcome depends on the value of another predictor. In that sense, Application 6 can be viewed as a special case of Application 7 where the grouping variable is the effect modifier.

The key conceptual distinction made by Vittinghoff et al. (2012) — but not explicitly highlighted by Kleinbaum et al. (2014) — is between **prediction** (forecasting future outcomes) and **causal inference** (estimating the effect of a specific predictor). This distinction has important implications for model building strategy: prediction models can include any variables that improve predictive accuracy, while causal inference requires careful consideration of confounding, mediation, and the causal structure of the data.

6.1.2 Prediction, Inference, and Causal Inference

The goal of **prediction** is to forecast an outcome Y for new observations, given observed predictor variables $X_1, \ldots, X_p$. In a prediction problem, the model is often treated as a "black box": we care about predictive accuracy, not about the exact form or interpretation of the relationship between predictors and outcome (James et al. 2021, chap. 2).

Exm

Example 6.1. For example, a clinical prediction model might identify which patients are at highest risk of hospital readmission, using demographics, diagnoses, and laboratory values, without requiring that each predictor be causally related to readmission.

At its core, prediction aspires to minimize uncertainty: the goal is that, by including enough informative predictors and combining them flexibly, we can reduce as much as possible the residual randomness in our forecasts.

Statistical inference also studies the relationship between Y and $X_1, \ldots, X_p$, but the goal shifts from accuracy to understanding: which predictors are associated with the response, in what direction, and by how much (James et al. 2021, chap. 2)?

Exm

Example 6.2. For example, a cohort study might estimate the association between smoking and lung cancer incidence, along with a confidence interval quantifying uncertainty about the magnitude of that association.

Unlike prediction, statistical inference treats the uncertainty in outcomes as a meaningful quantity to characterize, not merely residual noise to be minimized.

Causal inference asks a distinct third question. A predictive association between a predictor X and an outcome Y does not imply that intervening to change X will change Y. Causal inference addresses this gap: it asks what would happen to Y if we were to actively *set* X to a particular value, rather than passively observe it (Vittinghoff et al. 2012, chap. 12).

Exm

Example 6.3. For example, even if smoking is strongly associated with lung cancer in observational data, causal inference asks whether reducing smoking rates would *cause* a reduction in lung cancer incidence — a question that requires additional assumptions about the data-generating process.

One useful framing of causal inference is as a prediction problem in **counterfactual** worlds: predicting what outcomes would have occurred under treatment conditions that were not actually administered. Under the **potential outcomes framework** (Neyman 1990; Rubin 1974, 688–701; Vittinghoff et al. 2012, chap. 12), individual causal effects are treated as deterministic in principle — each person has a fixed potential outcome under each treatment condition — and uncertainty arises from the fact that we can observe only one potential outcome per person, and from heterogeneity in effects across individuals.

Accurate causal inference typically requires accounting for **confounding**: in observational data, people who receive a treatment are often systematically different from those who do not in ways that also affect the outcome. These differences can make predictive associations misleading for causal questions.

A useful informal distinction: *prediction* asks which individuals are at highest risk, while *causal inference* asks what is the effect of a particular intervention, and *statistical inference* asks what the association is without necessarily specifying an intervention.

For an accessible introduction to the distinction between prediction and statistical inference, see James et al. (2021, chap. 2). For coverage of causal inference methods in an epidemiological setting, see Vittinghoff et al. (2012, chap. 12) and Hernán and Robins (2020, pt. I). For an in-depth treatment, see the causal inference chapter[4].

[4] causal-inference.qmd

6.2 Example: Adelie penguins

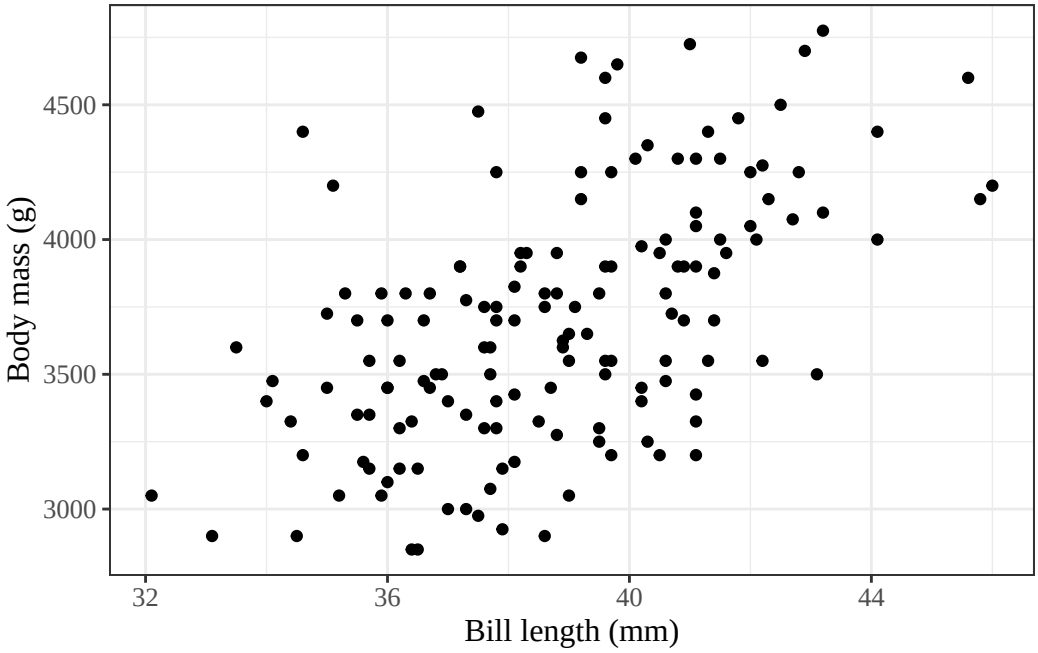


Figure 1: Palmer penguins

6.3 Linear regression

```
ggpenguins2 <-
  ggpenguins +
  stat_smooth(
    method = "lm",
    formula = y ~ x,
    geom = "smooth"
  )

ggpenguins2 |> print()
```

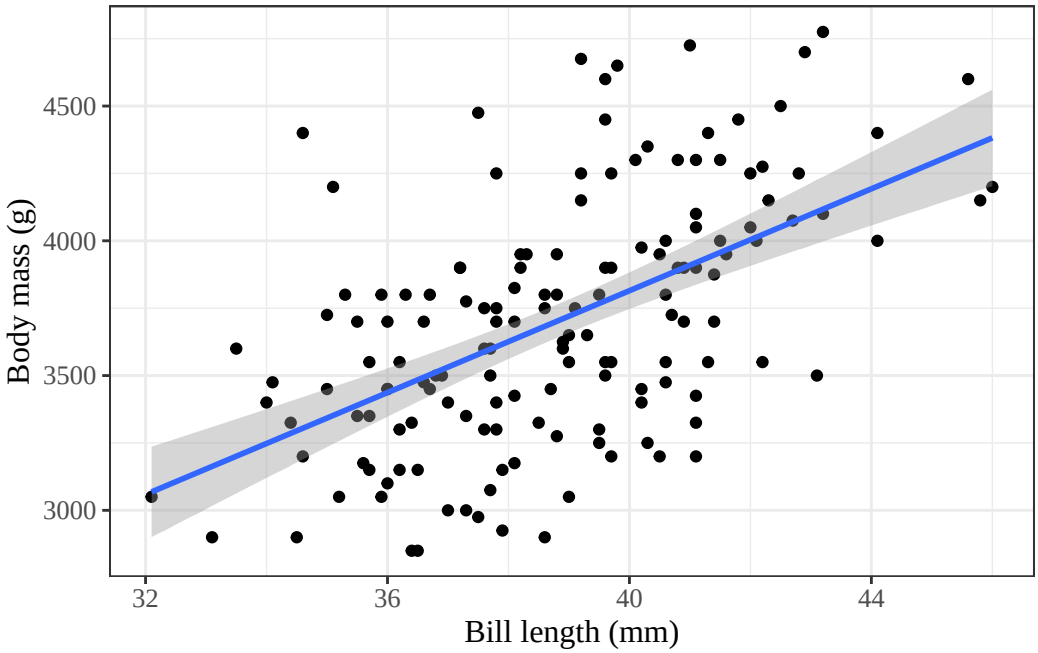


Figure 2: Palmer penguins with linear regression fit

6.4 Curved regression lines

```
ggpenguins2 <- ggpenguins +
  stat_smooth(
    method = "lm",
    formula = y ~ log(x),
    geom = "smooth"
  ) +
  xlab("Bill length (mm)") +
  ylab("Body mass (g)")
ggpenguins2
```

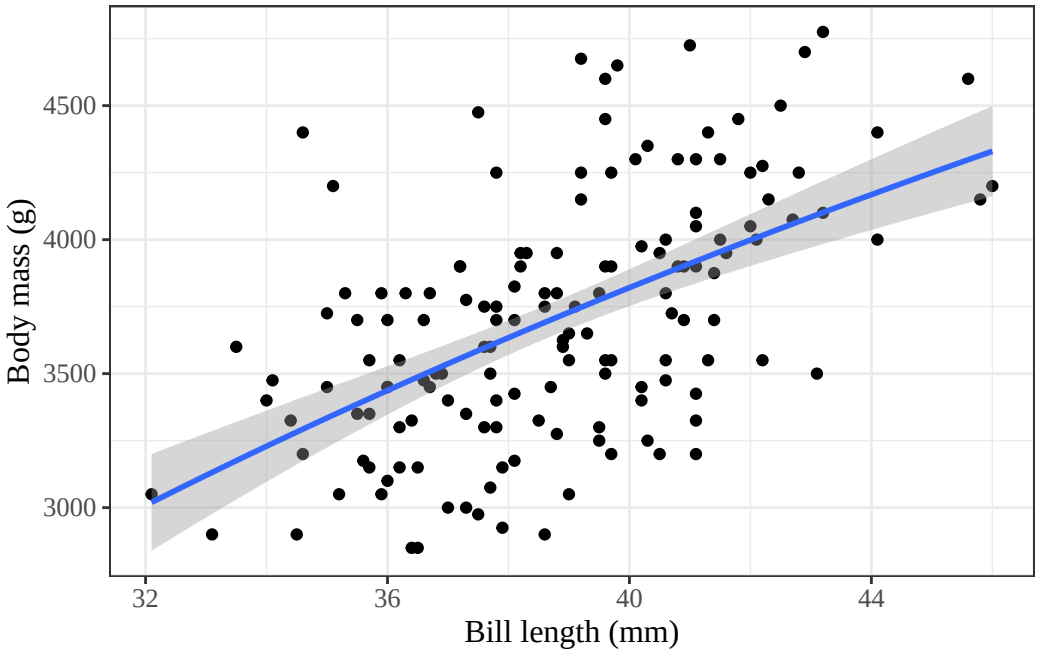


Figure 3: Palmer penguins - curved regression lines

6.5 Multiple regression

```
ggpenguins <-
  palmerpenguins::penguins |>
  ggplot(
    aes(
      x = bill_length_mm,
      y = body_mass_g,
      color = species
    )
  ) +
  geom_point() +
  stat_smooth(
    method = "lm",
    formula = y ~ x,
    geom = "smooth"
  ) +
  xlab("Bill length (mm)") +
  ylab("Body mass (g)")
ggpenguins |> print()
```

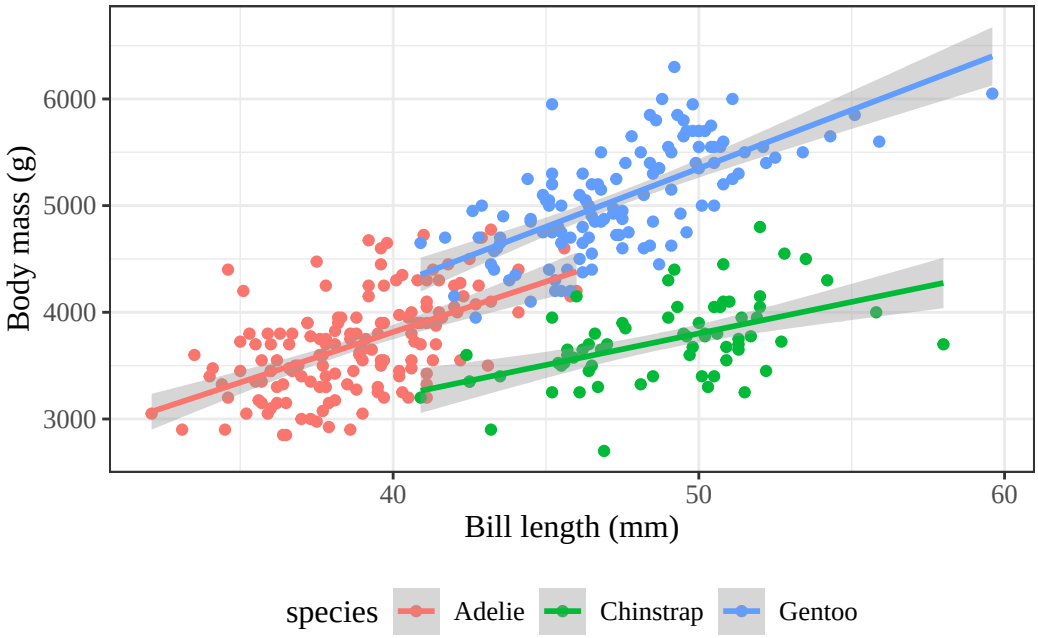


Figure 4: Palmer penguins - multiple groups

6.6 Modeling non-Gaussian outcomes

```
library(glmx)
data(BeetleMortality)
beetles <- BeetleMortality |>
  mutate(
    pct = died / n,
    survived = n - died
  )

plot1 <-
  beetles |>
  ggplot(aes(x = dose, y = pct)) +
  geom_point(aes(size = n)) +
  xlab("Dose (log mg/L)") +
  ylab("Mortality rate (%)") +
  scale_y_continuous(labels = scales::percent) +
  # xlab(bquote(log[10]), bquote(CS[2])) +
  scale_size(range = c(1, 2))

print(plot1)
```

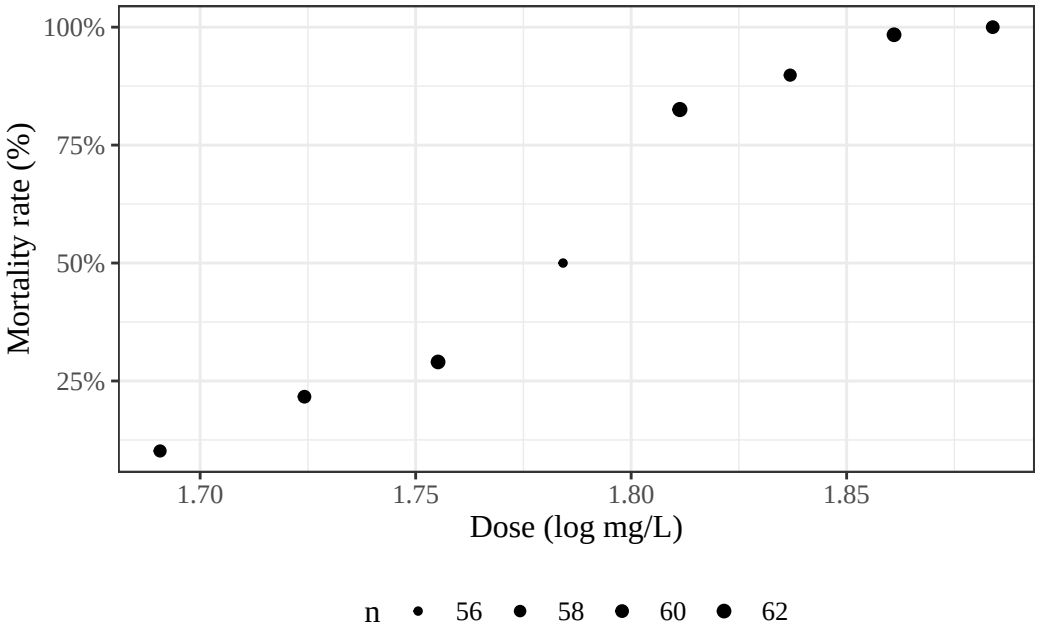


Figure 5: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

6.7 Why don't we use linear regression?

```
beetles_long <-
  beetles |>
  reframe(
    .by = everything(),
    outcome = c(
      rep(1, times = died),
      rep(0, times = survived)
    )
  )

lm1 <-
  beetles_long |>
  lm(
    formula = outcome ~ dose,
    data = _
  )


range1 <- range(beetles$dose) + c(-.2, .2)

f_linear <- function(x) predict(lm1, newdata = data.frame(dose = x))

plot2 <-
  plot1 +
  geom_function(fun = f_linear, aes(col = "Straight line")) +
  labs(colour = "Model", size = "")
print(plot2)
```

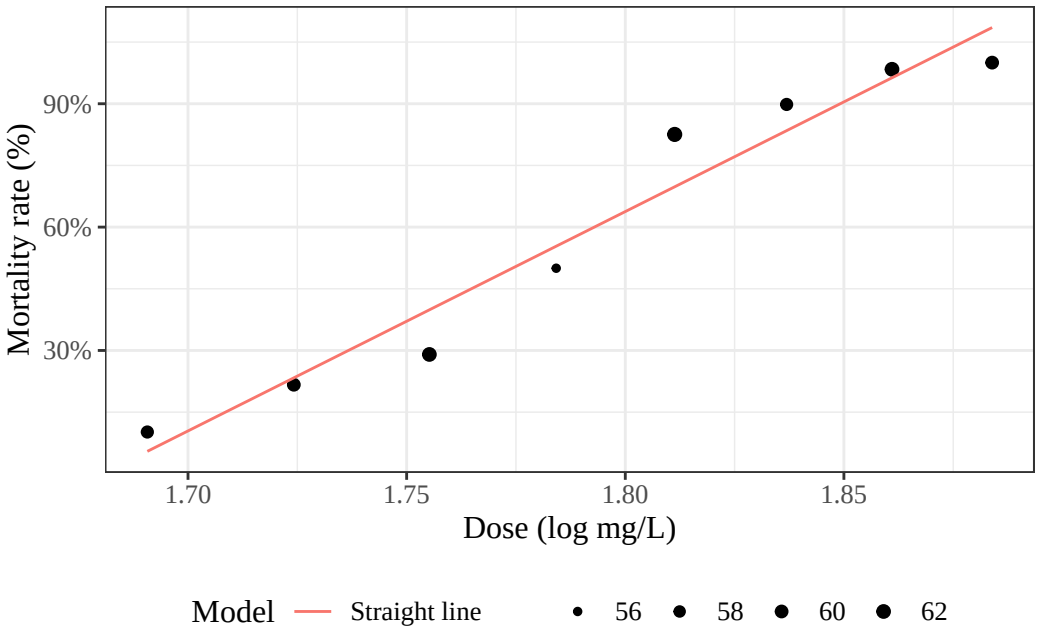


Figure 6: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

6.8 Zoom out

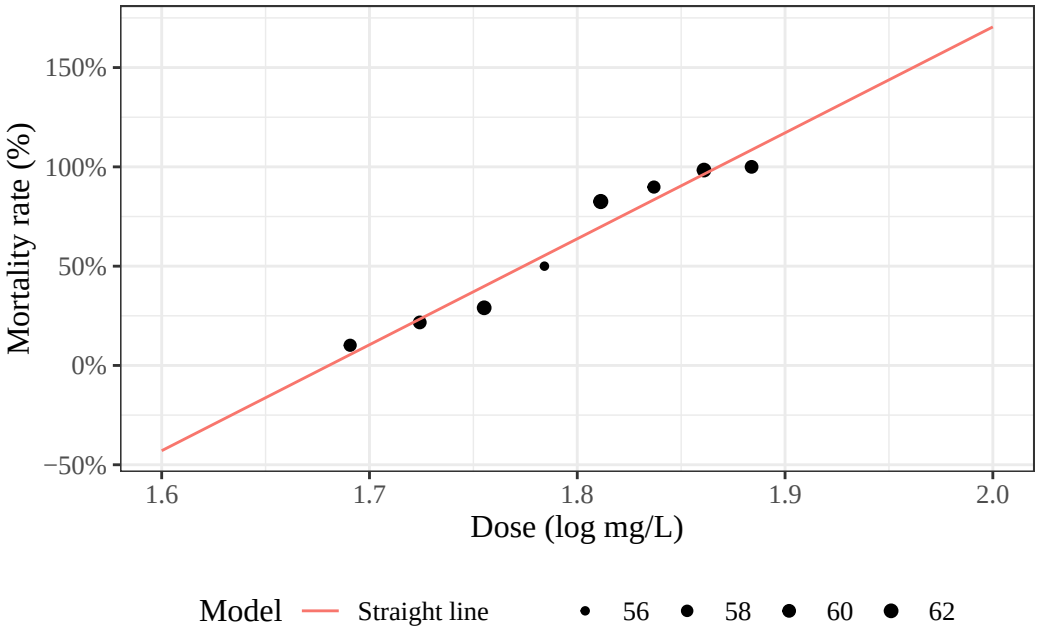


Figure 7: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

6.9 log transformation of dose?

```
lm2 <-
  beetles_long |>
  lm(formula = outcome ~ log(dose), data = _)

f_linearlog <- function(x) predict(lm2, newdata = data.frame(dose = x))

plot3 <- plot2 +
  expand_limits(x = c(1.6, 2)) +
  geom_function(fun = f_linearlog, aes(col = "Log-transform dose"))

print(plot3 + expand_limits(x = c(1.6, 2)))
```

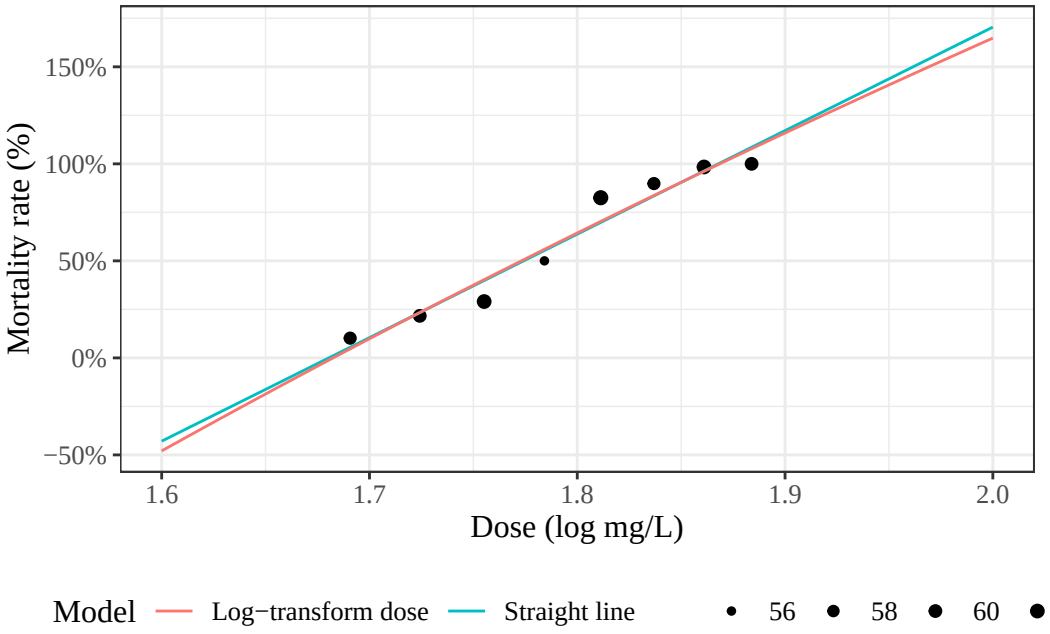


Figure 8: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

6.10 Logistic regression

```
glm1 <- beetles |>
  glm(formula = cbind(died, survived) ~ dose, family = "binomial")

f <- function(x) {
  glm1 |>
    predict(newdata = data.frame(dose = x), type = "response")
}

plot4 <- plot3 + geom_function(fun = f, aes(col = "Logistic regression"))
print(plot4)
```

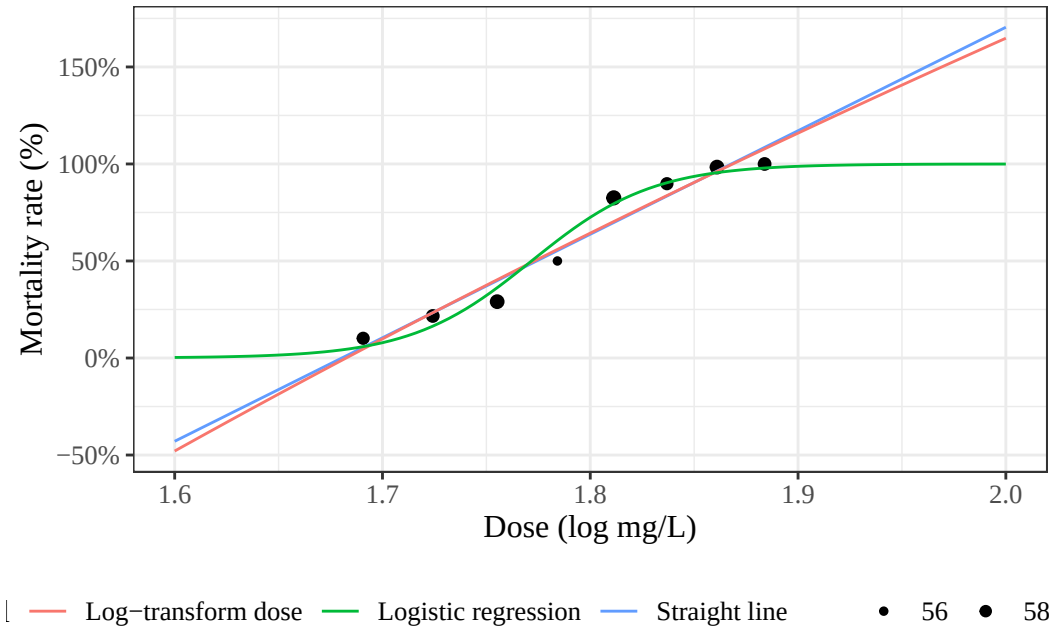


Figure 9: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

7 Structure of regression models

Exercise 7.1. What is a regression model?

Definition 7.1 (Regression model)**.** Regression models are conditional probability distribution models:

$$\mathrm{P}(Y|\tilde{X})$$

Exercise 7.2. What are some of the names used for the variables in a regression model $\mathrm{P}(Y|\tilde{X})$?

Definition 7.2 (Outcome)**.** The outcome variable in a regression model is the variable whose distribution is being described; in other words, the variable on the left-hand side of the "|" ("pipe") symbol.
The outcome variable is also called the **response variable**, **regressand**, **predicted variable**, **explained variable**, **experimental variable**, **output variable**, **dependent variable**, **endogenous variables**, **target**, or **label**.
and is typically denoted Y.

Definition 7.3 (Predictors)**.** The predictor variables in a regression model are the conditioning variables defining subpopulations among which the outcome distribution might vary.
Predictors are also called **regressors**, **covariates**, **independent variables**, **explanatory variables**, **risk factors**, **exposure variables**, **input variables**, **exogenous variables**, **candidate variables** (Dunn and Smyth (2018)), **carriers** (Dunn and Smyth (2018)), **manipulated variables**, or **features** and are typically denoted $\tilde{X}$. [5]

Table 3: Common pairings of terms for variables $\tilde{X}$ and Y in regression models $P(Y|\tilde{X})$ [7]

$\tilde{X}$	Y	usual context
input	output	
independent	dependent	
predictor	predicted or response	
explanatory	explained	
exogenous	endogenous	econometrics
manipulated	measured	randomized controlled experiments
exposure	outcome	epidemiology
feature	label or target	machine learning

Exercise 7.3. What is the general structure of a generalized linear model?

Solution

Solution 7.1. Generalized linear models have three components:

1. The **outcome distribution** family: $\mathrm{p}(Y|\mu(\tilde{x}))$
2. The **link function**: $g(\mu(\tilde{x})) = \eta(\tilde{x})$
3. The **linear component**: $\eta(\tilde{x}) = \tilde{x} \cdot \beta$

---

1. The **outcome distribution** family (a.k.a. the **random component** of the model)
- Gaussian (normal)
- Binomial
- Poisson
- Exponential
- Gamma
- Negative binomial

---

2. The **linear component** (a.k.a. the *linear predictor* or *linear functional form*) describing how the covariates combine to define subpopulations:

[5] The "~" ("tilde") symbol in the notation $\tilde{X}$ indicates that $\tilde{X}$ is a vector. See the notation page of the lab's math notes[6] for a table of notation used in these notes.
[7] adapted from https://en.wikipedia.org/wiki/Dependent_and_independent_variables#Synonyms

$$\eta(\tilde{x}) \stackrel{\text{def}}{=} \tilde{x}^\top \tilde{\beta} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ...$$

---

3. The **link function** relating the outcome distribution to the linear component, typically through the mean:

- identity: $\mu(y) = \eta(\tilde{x})$
- logit: $\log\left\{\frac{\mu(y)}{1-\mu(y)}\right\} = \eta(\tilde{x})$
- log: $\log\{\mu(y)\} = \eta(\tilde{x})$
- inverse: $(\mu(y))^{-1} = \eta(\tilde{x})$
- clog-log: $\log\{-\log\{1-\mu(y)\}\} = \eta(\tilde{x})$

Components 2 and 3 together are sometimes called the **systematic component** of the model (for example, in Dunn and Smyth (2018)).

8 Why link functions?

So far, we have seen models for Gaussian distributions that vary across subpopulations, and distributions for single populations. A GLM combines the two: it describes how a non-Gaussian distribution varies across subpopulations. The link function is the piece that makes this work.

Exercise 8.1. Suppose the outcome is a count or a binary variable. Why not model the mean $\mu(\tilde{x})$ as a linear function of the covariates, as in a Gaussian model?

Solution

Solution 8.1. A straight line for $\mu(\tilde{x})$ can leave the range of the outcome: it can predict a negative count, or a probability below 0 or above 1. We could try to repair this by transforming each covariate separately and adding interaction terms until the fitted curve stays in range, but that is hard to get right.

The link function offers a simpler route. We add all the covariate terms together first, to get the linear predictor $\eta(\tilde{x})$, and then transform once, with the inverse link g^{-1}. The link g relates the mean to the linear predictor: $g(\mu(\tilde{x})) = \eta(\tilde{x})$. Applying g^{-1} to both sides gives:

$$\mu(\tilde{x}) = g^{-1}(\eta(\tilde{x}))$$

Choosing the inverse link guarantees the model structure we want:

- with the log link, $\mu(\tilde{x}) = \exp\{\eta(\tilde{x})\} > 0$, so a mean count is never negative;
- with the logit link, $\mu(\tilde{x}) = \text{expit}(\eta(\tilde{x}))$ lies in $(0, 1)$, so a probability is never below 0 or above 1.

Remark 8.1 (Slopes under a non-identity link). With a non-identity link, interpreting a slope gets harder. A coefficient β_j is the change in $g(\mu)$ per unit change in x_j, not the change in μ. Even without any interaction terms, the change in μ per unit change in x_j depends on the values of the other covariates, because g^{-1} is not linear.

Exm

Example 8.1 (A logistic model with no interaction)**.** Suppose the mean μ of a binary outcome, which is its probability, satisfies $\text{logit}(\mu) = -2 + 1 \cdot x + 1.5 \cdot z$, with $z \in \{0, 1\}$. The coefficient of x is the same at both values of z, so the odds ratio for a one-unit increase in x is e^1 at both values of z. The code below computes the change in probability for the same increase in x, from $x = 0$ to $x = 1$.

```r
risk_diff <- function(z) {
  plogis(-2 + 1 * 1 + 1.5 * z) - plogis(-2 + 1 * 0 + 1.5 * z)
}
rd0 <- risk_diff(0)
rd1 <- risk_diff(1)
```

For $z = 0$, the probability rises by 0.15; for $z = 1$, it rises by 0.245. The odds ratio is the same in both subgroups, but the difference in probabilities is not.

9 Canonical links

The link functions listed in the structure of a GLM are not all on the same footing. For each outcome distribution, the form of its density singles out one link. To say which, we need the density's *natural parameter*.

Definition 9.1 (Exponential dispersion family, natural parameter, and canonical link)**.** A distribution belongs to the **exponential dispersion family** if its density (or probability mass function) can be written as

$$f(y; \theta, \phi) = \exp\left\{ \frac{y\,\theta - b(\theta)}{\phi} + c(y, \phi) \right\}$$

for some functions $b(\cdot)$ and $c(\cdot, \cdot)$. The parameter θ is the **natural parameter**, and $\phi > 0$ is the **dispersion parameter**. Then

$$\begin{aligned} \mathrm{E}[Y] &= b'(\theta) \stackrel{\text{def}}{=} \mu \\ \mathrm{Var}(Y) &= \phi\, b''(\theta) = \phi\, V(\mu), \end{aligned}$$

where $V(\mu) \stackrel{\text{def}}{=} b''(\theta)$, written as a function of μ, is the **variance function**. The **canonical link** is the link function $g(\cdot)$ that sets the linear predictor equal to the natural parameter: $g(\mu) \stackrel{\text{def}}{=} \theta$ (Dobson and Barnett 2018, chap. 3; Dunn and Smyth 2018, chap. 5).

Exm

Example 9.1 (Natural parameters of a Poisson and a Bernoulli outcome)**.** For a Poisson outcome with mean μ,

$$f(y; \mu) = \frac{\mu^y e^{-\mu}}{y!} = \exp\{y \log\{\mu\} - \mu - \log\{y!\}\},$$

so $\theta \stackrel{\text{def}}{=} \log\{\mu\}$, $b(\theta) \stackrel{\text{def}}{=} e^\theta$, and $\phi \stackrel{\text{def}}{=} 1$. For a Bernoulli outcome with success probability π,

$$f(y; \pi) = \pi^y (1 - \pi)^{1-y} = \exp\left\{ y \log\left\{ \frac{\pi}{1 - \pi} \right\} + \log\{1 - \pi\} \right\},$$

so $\theta \stackrel{\text{def}}{=} \mathrm{logit}(\pi)$, $b(\theta) \stackrel{\text{def}}{=} \log\{1 + e^\theta\}$, and $\phi \stackrel{\text{def}}{=} 1$.
The code below checks both forms against R's built-in densities, at $\mu = 3$, $y = 2$ and at $\pi = 0.25$, $y = 1$, and computes $b'(\theta)$ and $b''(\theta)$ for the Bernoulli case.

```r
mu <- 3
theta_pois <- log(mu)
y_pois <- 2
pois_exp_form <- exp(y_pois * theta_pois - exp(theta_pois) - lgamma(y_pois + 1))
pois_check <- isTRUE(all.equal(pois_exp_form, dpois(y_pois, mu)))

pi_bern <- 0.25
theta_bern <- qlogis(pi_bern)
y_bern <- 1
bern_exp_form <- exp(y_bern * theta_bern - log1p(exp(theta_bern)))
bern_check <- isTRUE(all.equal(bern_exp_form, dbinom(y_bern, 1, pi_bern)))

b1_bern <- plogis(theta_bern)
b2_bern <- b1_bern * (1 - b1_bern)
```

- For the Poisson distribution with $\mu = 3$, the natural parameter is $\theta = \log(3) = 1.0986$, and the exponential-family form gives $f(2) = 0.224$, which matches `dpois(2, 3)`: TRUE. Here $b'(\theta) = b''(\theta) = e^{\theta} = 3$, so $V(\mu) = \mu$.
- For the Bernoulli distribution with $\pi = 0.25$, the natural parameter is $\theta = \text{logit}(0.25) =$ -1.0986, and the exponential-family form gives $f(1) = 0.25$, which matches `dbinom(1, 1, 0.25)`: TRUE. Here $b'(\theta) = \text{expit}(\theta) = 0.25$, which is π, and $b''(\theta) = \pi(1-\pi) = 0.1875$, so $V(\pi) = \pi(1-\pi)$.

Consequently, the canonical link for the Poisson distribution is the log link, and the canonical link for the Bernoulli distribution is the logit link.

Exercises

Exercise 9.1 (Identifying GLM components)**.** (adapted from Dobson and Barnett (2018), Chapter 4, and Dunn and Smyth (2018), Chapter 5)
For each of the following scenarios, identify the three components of the GLM: (1) the outcome distribution, (2) the link function $g(\cdot)$, and (3) the linear predictor $\eta = \tilde{x} \cdot \tilde{\beta}$.
(a) Modeling the probability of a binary outcome (disease/no disease) as a function of continuous predictors, using the logit link.
(b) Modeling count data (number of events per person-year) as a function of continuous predictors, using the log link.
(c) Modeling a continuous, strictly positive outcome using a gamma distribution and log link.

Solution

Solution. **(a)** Logistic regression:
1. **Outcome distribution**: $Y_i \mid \tilde{x}_i \sim_{\perp\!\!\!\perp} \text{Bernoulli}(\pi_i)$
2. **Link function**: logit link, $g(\pi) = \log\{\pi/(1-\pi)\}$
3. **Linear predictor**: $\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$

The mean function (inverse link) is: $\pi_i = \text{expit}(\eta_i) = \frac{e^{\eta_i}}{1+e^{\eta_i}}$.
(b) Poisson (log-linear) regression:
1. **Outcome distribution**: $Y_i \mid \tilde{x}_i \sim_{\perp\!\!\!\perp} \text{Pois}(\mu_i)$
2. **Link function**: log link, $g(\mu) = \log\{\mu\}$
3. **Linear predictor**: $\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$

The mean function (inverse link) is: $\mu_i = e^{\eta_i}$.
(c) Gamma regression with log link:
1. **Outcome distribution**: $Y_i \mid \tilde{x}_i \sim_{\perp\!\!\!\perp} \text{Gamma}(\text{mean} = \mu_i)$
2. **Link function**: log link, $g(\mu) = \log\{\mu\}$
3. **Linear predictor**: $\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$

The mean function is: $\mu_i = e^{\eta_i}$.

Exercise 9.2 (Canonical link functions)**.** (adapted from Dobson and Barnett (2018), Chapter 3, and Dunn and Smyth (2018), Chapter 5)
The *canonical link* for a distribution in the exponential family is the link $g(\cdot)$ such that $g(\mu) = \theta$, where θ is the natural parameter (see Canonical links[a]).
For each listed distribution, state the canonical link function and the corresponding variance function $V(\mu)$, where $\text{Var}(Y) = \phi\, V(\mu)$ and ϕ is the dispersion parameter:
(a) $Y \sim \text{Pois}(\mu)$
(b) $Y \sim \text{Bernoulli}(\pi)$
(c) $Y \sim \text{N}(\mu, \sigma^2)$
(d) $Y \sim \text{Gamma}(\mu, \phi)$, with mean μ and shape $1/\phi$

[a] Intro-to-GLMs.qmd#sec-canonical-link

Solution

Solution.

Distribution	Canonical link	Variance function $V(\mu)$	$\text{Var}(Y) = \phi V(\mu)$
(a) Poisson	$g(\mu) = \log\{\mu\}$	$V(\mu) = \mu$	$\phi = 1$ (fixed)
(b) Bernoulli	$g(\pi) = \log\{\frac{\pi}{1-\pi}\}$ (logit)	$V(\pi) = \pi(1-\pi)$	$\phi = 1$ (fixed)
(c) Normal	$g(\mu) = \mu$ (identity)	$V(\mu) = 1$	$\text{Var}(Y) = \sigma^2$
(d) Gamma	$g(\mu) = -1/\mu$ (inverse)	$V(\mu) = \mu^2$	$\text{Var}(Y) = \phi\mu^2$

Notes:

- For the Poisson distribution, the variance equals the mean. Overdispersion occurs when the empirical variance exceeds the mean.
- For the Bernoulli distribution, the variance is maximized at $\pi = 0.5$.
- For the Normal distribution, the variance does not depend on the mean; the dispersion parameter $\phi = \sigma^2$.
- For the Gamma distribution, the coefficient of variation $\text{SD}(Y)/\,\text{E}[Y]$ is constant.
- The sign of the Gamma canonical link is a convention: $\theta = -1/\mu$ in the density's natural parameterization, while R's `Gamma()` family uses $1/\mu$ as its default inverse link. Either way, the linear predictor can produce a negative μ.

The log link (not the canonical inverse link) is often used for Gamma regression in practice, because it ensures $\mu > 0$ and gives interpretable multiplicative effects.

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